mitrafield

Explain it once.
Every beat finds its engine.

An AI director that writes explainer videos and sends every beat to the engine that can show it truthfully: exact math, real maps, charts and diagrams where accuracy matters, Veo and Nano Banana where mood does.

Google Cloud AI Builder Cup 2026 · Theme: **Media, Content & Digital Experiences**

Viraj Sawad · github.com/VirajSawad1021/MitraField

THE PROBLEM

Explainer videos are how the world learns.
Making a correct one is still slow, expensive and manual.

days

of work per minute

A good explainer needs a script, research, animation, maps, charts, narration and editing. Teachers, NGOs and small creators cannot afford a studio.

wrong

numbers drawn by pixels

Text-to-video tools make beautiful clips, but they hallucinate. Coastlines are invented, figures are garbled, equations are squiggles. For learning, a wrong number is worse than none.

one

language, one format

Most explainers exist only in English and only as long-form. Re-making them as Shorts or in Hindi, Nepali or Bahasa means starting again.

THE SOLUTION

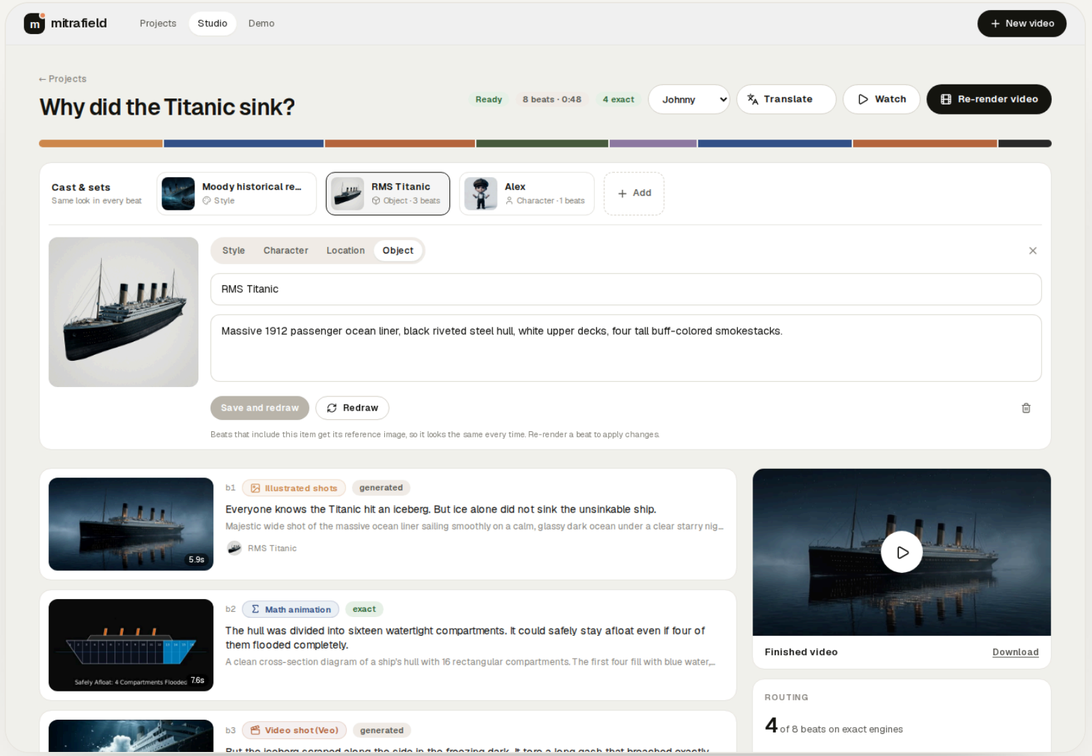
A director agent with ten engines.
Precise where it matters, generative where it helps.

You ask a question and add notes or PDFs. Gemini writes a storyboard for the ear, beat by beat, and routes each beat to the engine that can show it truthfully. Every beat is rendered to the exact length of its narration, checked, and assembled into a finished video with music and subtitles.

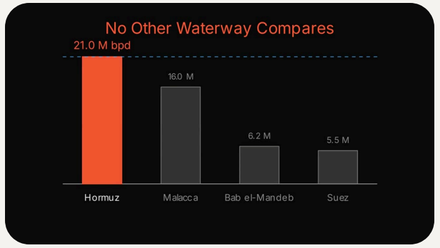
exact Math animation · Globe & maps · Data chart · Timeline · Flow diagram · Whiteboard · Kinetic type

generated Illustrated shots · Guide character · Veo video

Facts, numbers, maps and math are always drawn by code from real data. Generated pixels never carry a fact.



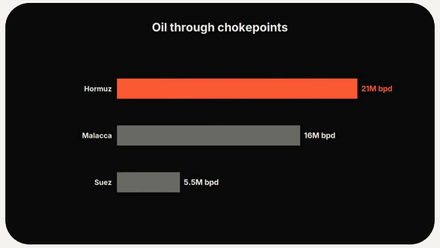
Every engine plans a beat and renders it to the exact length.



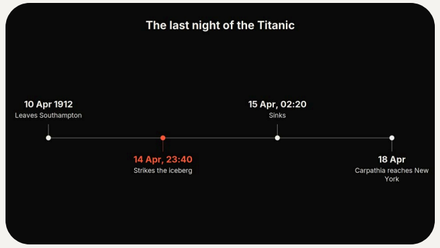
Math animation
Gemini writes Manim code, repairs errors, checks frames



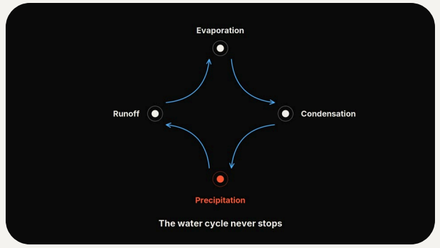
Globe & maps
Natural Earth borders, flights, pins, routes



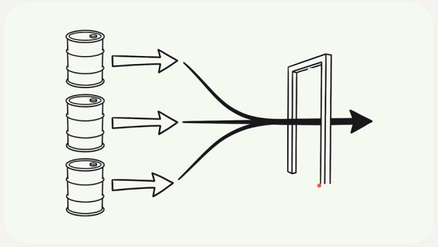
Data chart
Bars and trends from source numbers



Timeline
Events at their real dates



Flow diagram
Processes, cycles, trees



Whiteboard
Line art drawn on stroke by stroke



Kinetic type
Hooks and key numbers



Illustrated shots
Nano Banana stills with camera moves



Guide character
One presenter, same look every beat



Veo shots
Hero motion, max two per video

HOW IT WORKS

From a question to a finished video.
You only make the calls.

01	**Ask**	YOU	A question plus optional notes, PDFs or images. Format, length, language, narrator, audience.
02	**Storyboard + cast**	GEMINI 3.1 PRO	Narration beat by beat, an engine per beat, and the recurring cast: style, characters, objects, places.
03	**Narrate**	GEMINI TTS	Each line is voiced first, then transcribed and retaken if it drifts from the script.
04	**Render + self-check**	ENGINES	Beats render in parallel to their exact length. Manim code is repaired on error and inspected by Gemini vision.
05	**Edit one beat**	YOU + DIRECTOR	Rewrite a line, swap an engine, or say "show it on a map". Only that beat re-renders.
06	**Assemble + translate**	LYRIA · FFMPEG	Music ducked under the voice, -14 LUFS, subtitles. One-click versions in other languages reuse the visuals.

Define it once. The same ship in every shot.

The director defines the recurring ingredients with the storyboard. Each gets one reference image; every generated beat is drawn from the references it contains, and Veo animates an on-model first frame.



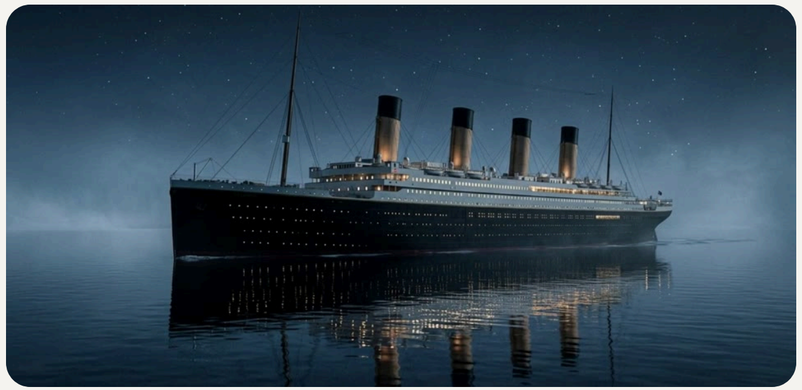
RMS Titanic
object · black hull, white decks, four buff funnels



Alex
guide character · 1912 newsboy cap



Moody historical realism
style · applied to every generated beat



Beat 1 · illustrated still



Beat 3 · Veo video



Beat 7 · Veo video



Beat 5 · guide character

ARCHITECTURE ON GOOGLE CLOUD

One Cloud Run service, Google models end to end.

Workspace
Next.js app: new video, studio, cast strip, inspector, projects, demo

Your sources
Notes, PDFs and images read by Gemini as the source of truth

→

Cloud Run · FastAPI

Director agent · per-beat pipeline (narrate → plan → render → fit) · content-hash cache · cost ledger · passcode for writes

Code engines
Manim + LaTeX, pycairo globe, charts, timeline, flow

Assembly
ffmpeg: fit, crossfade, mix, loudnorm, .srt

Cloud Storage mount for projects and media · Secret Manager for keys

→

Gemini 3.1 Pro
storyboard, routing, cast, rewrites, translation

Gemini Flash
engine planning, Manim code, vision QA, transcripts

Nano Banana
stills, characters, line art, cast references

Veo 3.1
hero shots, image-to-video from cast frames

Gemini TTS · Lyria
narration · topic-matched music bed

GEN AI IMPLEMENTATION

Not one prompt. Five loops that check each other.

01

Route

The director reads every engine's description and picks per beat, under rules: facts go to exact engines, Veo is capped at two.

02

Code + repair

Gemini writes Manim for the exact duration. Render errors go back with the traceback until it compiles.

03

Look + fix

Gemini vision inspects rendered frames for overlapping or cut-off text and asks for a corrected scene.

04

Listen + retake

Every narration line is transcribed and retaken if words were added, dropped or garbled.

05

Stay on model

Cast references feed every image and Veo first frame, so characters and subjects stay consistent.

Failures degrade gracefully: a refused Veo clip becomes a still, a failed engine becomes a title card, and the video still renders.

RESULTS FROM THE PROTOTYPE

Finished explainers from a single prompt.

4

demo videos across geography, math, biology (vertical Short) and history

4–8 min

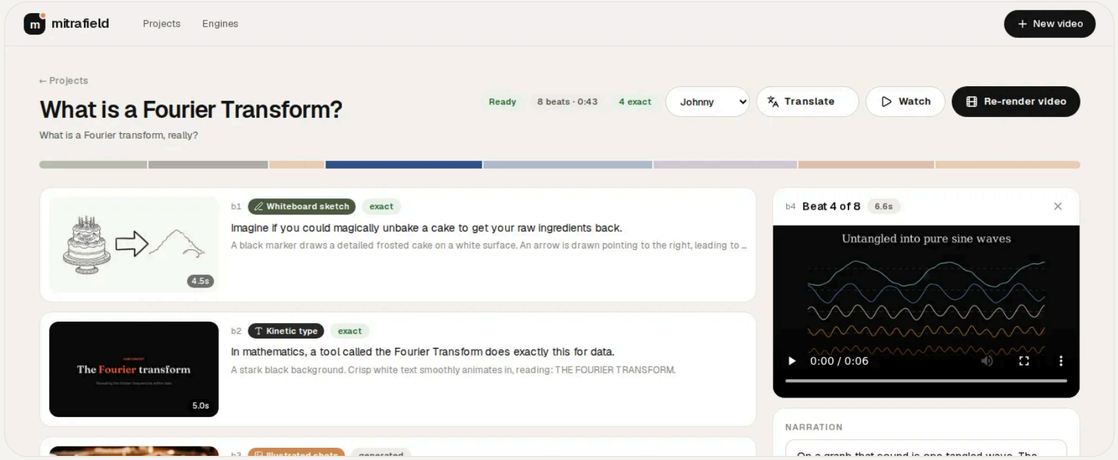
render time for a 45-50 s video on one small server, unattended

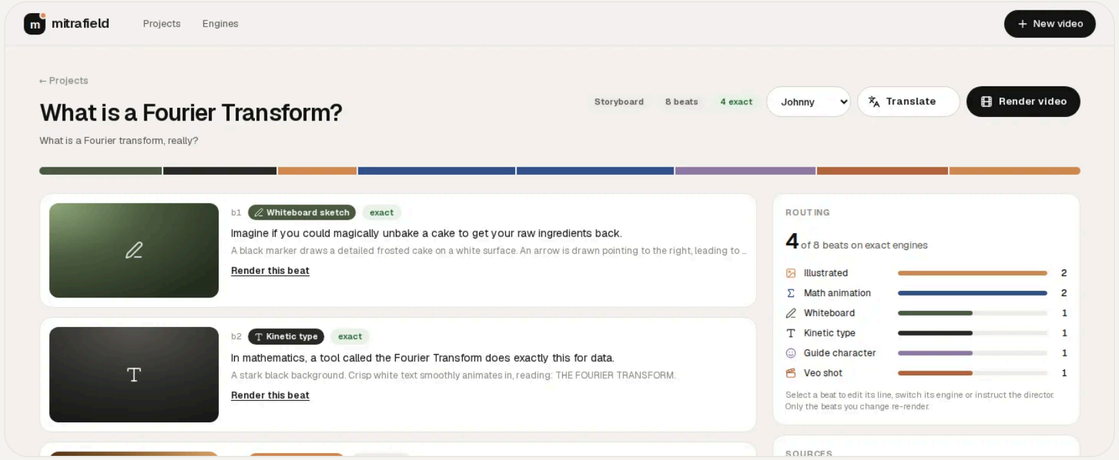
~$2

API cost per short video with two Veo shots; about $0.60 without Veo; draft mode halves Veo

50%

of beats on exact engines on average, so facts are computed, not generated





MitraField · costs from Google's published Gemini API rates, logged per call

IMPACT

Correct explainers for everyone who teaches.
In their language, in their format.

Teachers and schools

Turn a lesson plan into a narrated video with accurate diagrams, then into a Short for revision. One teacher, minutes, a few dollars.

NGOs and newsrooms

Explain a policy, a disease or a crisis with real maps and real numbers, cited on screen, and publish it in several JAPAC languages the same day.

Creators

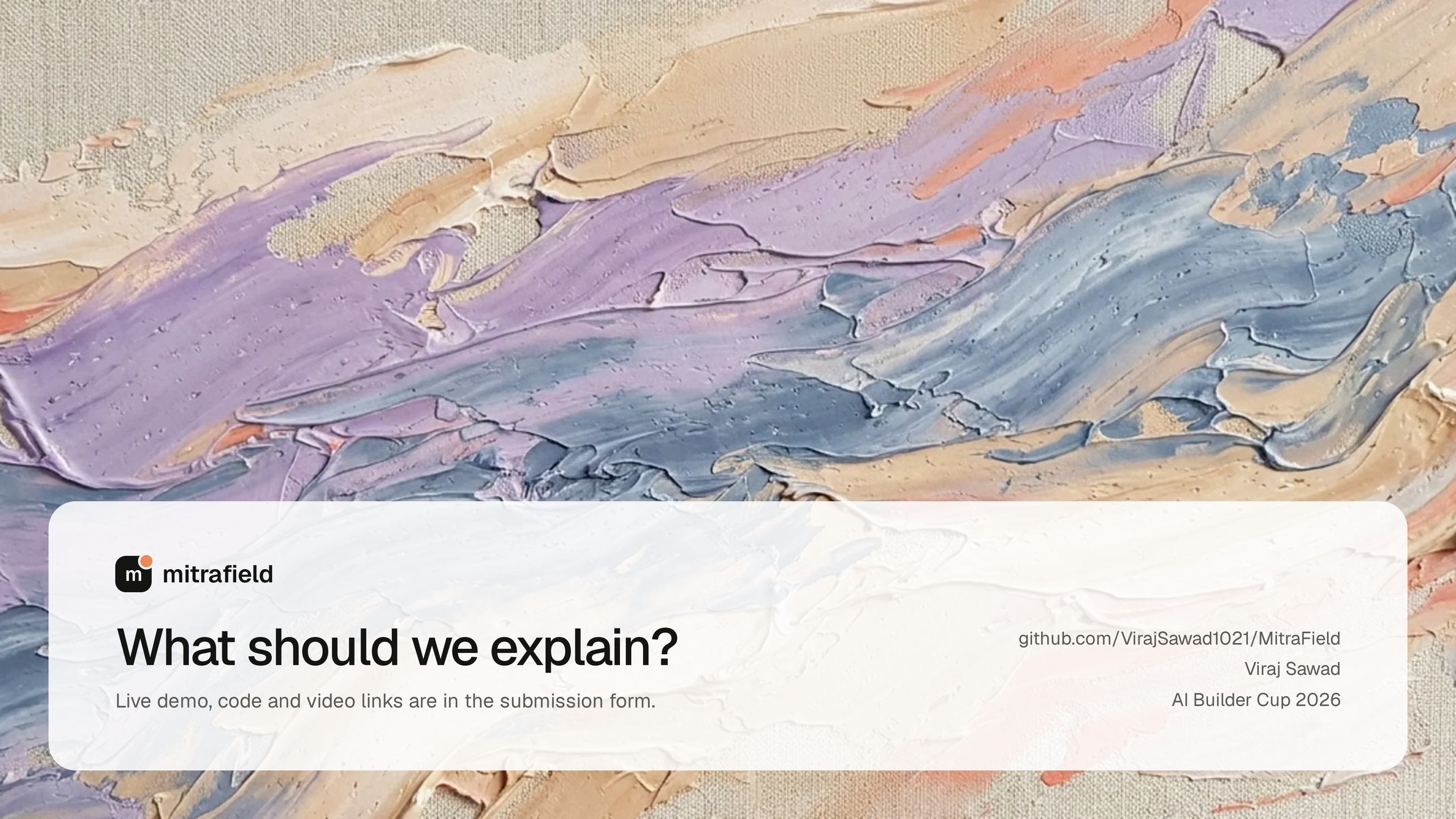
Run an explainer channel with a consistent presenter and style across every episode, long-form and Shorts from one workspace.

Misinformation angle: every number and map in a MitraField video is computed from data the user can see, not invented by a pixel model.

FEASIBILITY AND SCALE

Built to run cheaply, and to grow.

AREA	TODAY	NEXT
Compute	One Cloud Run service, renders as in-process jobs, content-hash cache	Renders as Cloud Run jobs, one per beat, scale to zero
Cost control	Per-project ledger, Veo length fitted to the beat, draft mode on Veo Lite	Batch API for final renders, per-user budgets
Styles	Ten engines behind one contract, discovered automatically	Scale comparisons, source-quote cards, code walkthroughs, real archival footage
Learning	Audience and level steer the script	Learn mode: quizzes after the video, beats re-explained where viewers struggle, analytics in BigQuery
Languages	One-click translation that reuses the visuals	Cloud Translation glossaries, regional voices



mitrafield

What should we explain?

Live demo, code and video links are in the submission form.

github.com/VirajSawad1021/MitraField

Viraj Sawad

AI Builder Cup 2026